

Ida Bagus Dwiweka Naratama

M.S. Candidate in Robotics, Kwangwoon University · Robotics & AI Researcher

Seoul, South Korea · dwiwekan@gmail.com · dwiwekan.github.io/portfolio · github.com/dwiwekan
· linkedin.com/in/dwiwekan

SUMMARY

Robotics and AI researcher building perception, spatial-reasoning and navigation systems for mobile robots, covering computer vision, multimodal and generative AI, 3D scene understanding, SLAM and autonomous navigation, from model development to deployment on real robots. Open to research and industry roles from Feb 2027.

EDUCATION

M.S. in Robotics, Kwangwoon University · Seoul, South Korea Feb 2025 – Feb 2027 (expected)
GPA 4.5 / 4.5

Advised by Prof. Junghyun Oh, Robotics & AI Lab

Bachelor's in Information Technology, Udayana University · Bali, Indonesia Aug 2019 – Apr 2023
GPA 3.93 / 4.0, cum laude

Thesis on extractive text summarization using a textual entailment approach based on deep learning

EXPERIENCE

Graduate Researcher, Robotics & AI Lab (RAIL), Kwangwoon University · Seoul, South Korea Feb 2025 – Present

- Research on AI-based perception, spatial reasoning and navigation for mobile robots.
- Built an uncertainty-aware object-goal navigation framework with ensemble semantic-map prediction and exploration planning, reaching a 64.0% success rate across 500 HM3D episodes and outperforming deterministic semantic baselines.
- Developed LatentSGC (graph neural networks + conditional flow matching) for generative 3D scene graph completion, raising node F1 from 0.245 to 0.432 and edge F1 from 0.089 to 0.255 over deterministic baselines.
- Reimplemented a task-driven open-set 3D scene graph pipeline and deployed it on an AgileX Scout Mini with ROS 2 and Nav2.
- Built reproducible simulation pipelines with NVIDIA Isaac Sim, ROS 2, SLAM Toolbox, AMCL and Nav2, where selected configurations reached 100% navigation success across repeated construction-site trials.

AI Engineer, PukulEnam.ai · Bali, Indonesia Jan 2024 – Sep 2024

- Built a Llama 3 virtual teaching assistant with edge deployment on an NVIDIA Jetson Nano, and LLM applications using retrieval-augmented generation (RAG) and the OpenAI API.
- Increased the frame rate of DICOM medical image sequences with FiNNiGAN and SRGAN super-resolution, outperforming interpolation.
- Built a custom U-Net for anomaly detection on ACDC 2017 (cross-entropy, focal, Dice and Lovász losses) and applied the OWAD framework to malware anomaly detection, benchmarked against random forest and decision tree models.

NLP Researcher (Professor's Assistant), Udayana University · Bali, Indonesia Jul 2022 – Mar 2023

- Adapted the SNLI natural-language-inference dataset to Indonesian.
- Built NLI models with BiLSTM, self-attention and multi-head attention using Stanza, FastText, GloVe, Word2Vec and BERT.

PUBLICATIONS

1. **Ida Bagus Dwiweka Naratama**, Junghyun Oh. "Permutation-Invariant Graph-Space Conformal Prediction for 3D Scene Graph Completion." *AAAI Conference on Artificial Intelligence (AAAI 2027)*. **Under review**.
2. **Ida Bagus Dwiweka Naratama**, I Made Putra Arya Winata, Sangcheol Oh, Yuri Song, Junghyun Oh. "Transforming Explicit Map Prediction into Uncertainty Exploration for Object Goal Navigation." *International Journal of Control, Automation, and Systems (IJCAS)*. **Under revision**.
3. **Ida Bagus Dwiweka Naratama**, I Made Putra Arya Winata, Junghyun Oh. "Latent Flow Matching for Uncertainty-Aware 3D Scene Graph Completion from Partial Robot Observations." *International Conference on Control, Automation and Systems (ICCAS 2026)*. **Accepted**.
4. **Ida Bagus Dwiweka Naratama**, I Made Putra Arya Winata, Yuri Song, Junghyun Oh. "VQ-LaMa: Discrete Semantic Map Inpainting for Efficient Exploration." *Korea Robotics Society Annual Conference (KRoC 2026)*, pp. 858–860, Feb 2026. **Published**.
5. **Ida Bagus Dwiweka Naratama**, Yuri Song, Junghyun Oh. "Integrated Simulation Framework for Construction-Site Quadruped Navigation." *Proceedings of the KICEM Annual Conference (KICEM 2025)*, Jeju, Korea, pp. 273–274, Nov 2025. **Published**.

6. I Made Putra Arya Winata, Donghyun Lee, **Ida Bagus Dwiweka Naratama**, Junghyun Oh. “Attention-Guided Semantic Navigation Learning from Pose, Goal Text, and Visual via Exponential Reward Functions.” *International Conference on Control, Automation and Systems (ICCAS 2025)*, Incheon, Korea, pp. 139–144, Nov 2025. **Published**.

7. I Made Putra Arya Winata, Donghyun Lee, **Ida Bagus Dwiweka Naratama**, Junghyun Oh. “Leveraging Large Language Model Assistance on Explicit Object Target Prediction and Navigation in Unseen Area.” *Journal of Institute of Control, Robotics and Systems (J. ICROS 2025)*, vol. 31, no. 8, pp. 888–893, Aug 2025. DOI 10.5302/J.ICROS.2025.25.0102. **Published**.

TEACHING & MENTORING

Machine Learning Path Mentor , Bangkit Academy · Indonesia	Feb 2023 – Jul 2024
• Mentored 75 students across three cohorts of 25 in the machine learning path. • Coordinated in-person instructor-led training sessions. • Recognised as “Study Squad Dynamo” and “Energetic Guide Mentor”.	
Machine Learning Mentor , Indonesia AI · Depok, Indonesia	Feb 2024 – Aug 2024
• Taught time-series forecasting (ARIMA, LSTM) across 12 classes for 20 participants, with one-on-one mentoring sessions.	

SELECTED PROJECTS

1. **Open-Set 3D Scene Graphs on a Mobile Robot** (Kwangwoon University). Reimplemented a task-driven open-set 3D scene graph pipeline and deployed it on an AgileX Scout Mini with ROS 2 and Nav2.
2. **Text Summarization with Textual Entailment** (Mar 2023). My undergraduate thesis, an extractive summarizer driven by a natural-language-inference model and served as a Flask web app.
3. **Tanam.in, a Gardening App with Plant Recognition** (Jul 2022). Android gardening-education app on Google Cloud. Transfer-learning models identify vegetables and fruits and detect tomato and chili leaf diseases.
4. **Information Retrieval Application** (Nov 2021). Desktop search engine over text documents combining Boolean queries, TF-IDF, Jaccard, n-grams, cosine distance and the extended Boolean model.
5. **Computer Vision Toolkit** (Nov 2021). Desktop image-processing app for thresholding, morphology, Sobel, Laplacian and Canny edge detection, plus line and circle detection.
6. **Food Sentiment Analysis with the Twitter API** (Jun 2021). End-to-end big-data pipeline that collects tweets about popular Indonesian dishes via the Twitter API, stores them in MongoDB, scores them with TextBlob and visualises the results in Tableau.

Research projects with abstracts and figures are at <https://dwiwekan.github.io/portfolio/#projects>

SKILLS

Machine learning	PyTorch, Transformers, LLMs, VLMs, Generative modelling, Computer vision
Robotics	ROS 2, Nav2, SLAM, AMCL, Robot perception, Autonomous navigation
Simulation	NVIDIA Isaac Sim, Gazebo, Habitat
Programming	Python, C++, SQL
Tools	Git, Linux, Docker, CUDA
Languages	English (professional), Indonesian (native)